how a scientific paper is built

prof. dr. nicolas meseth

1 one shape for all papers
2 what goes where
3 the most common mistake
4 the abstract

readers don't read a paper from start to end, they jump to the room they need, so every room has to be where they expect it.

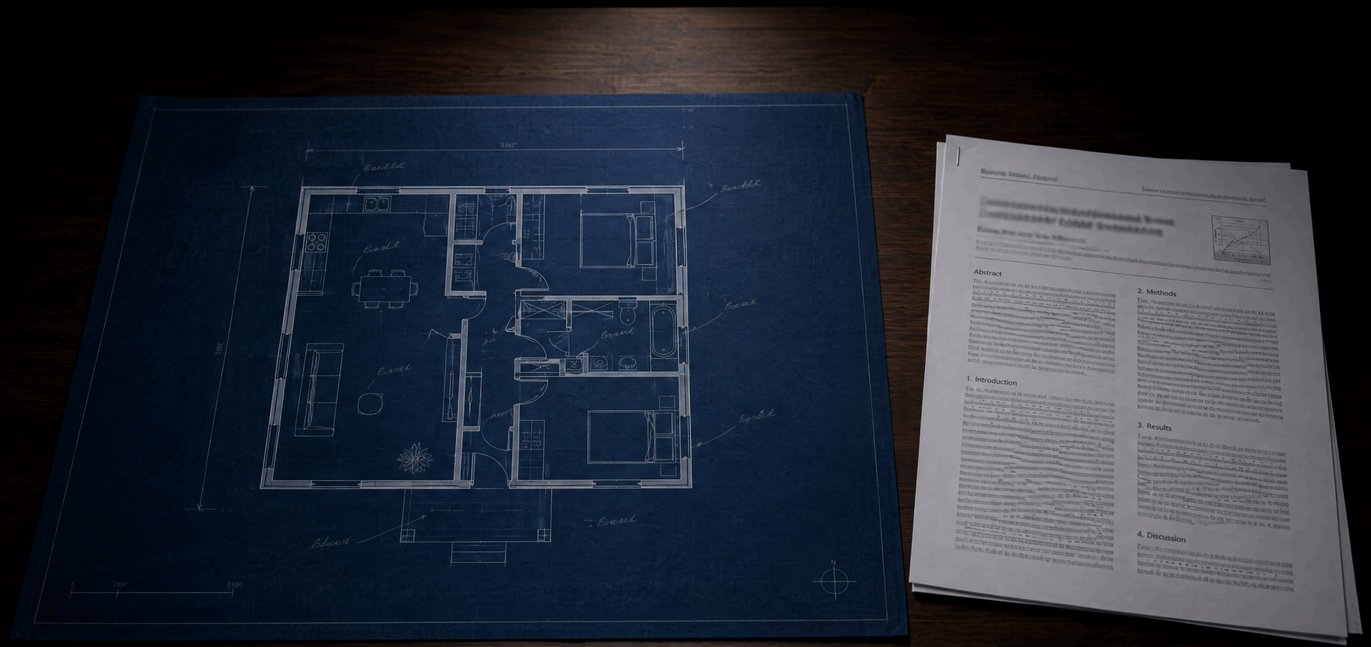


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part 1

one shape for all papers

imrad and its proportions

title, abstract

introduction

why, and what is the question?

references

title, abstract

introduction — why, and what is the question?

methods — how did we find out?

references

title, abstract

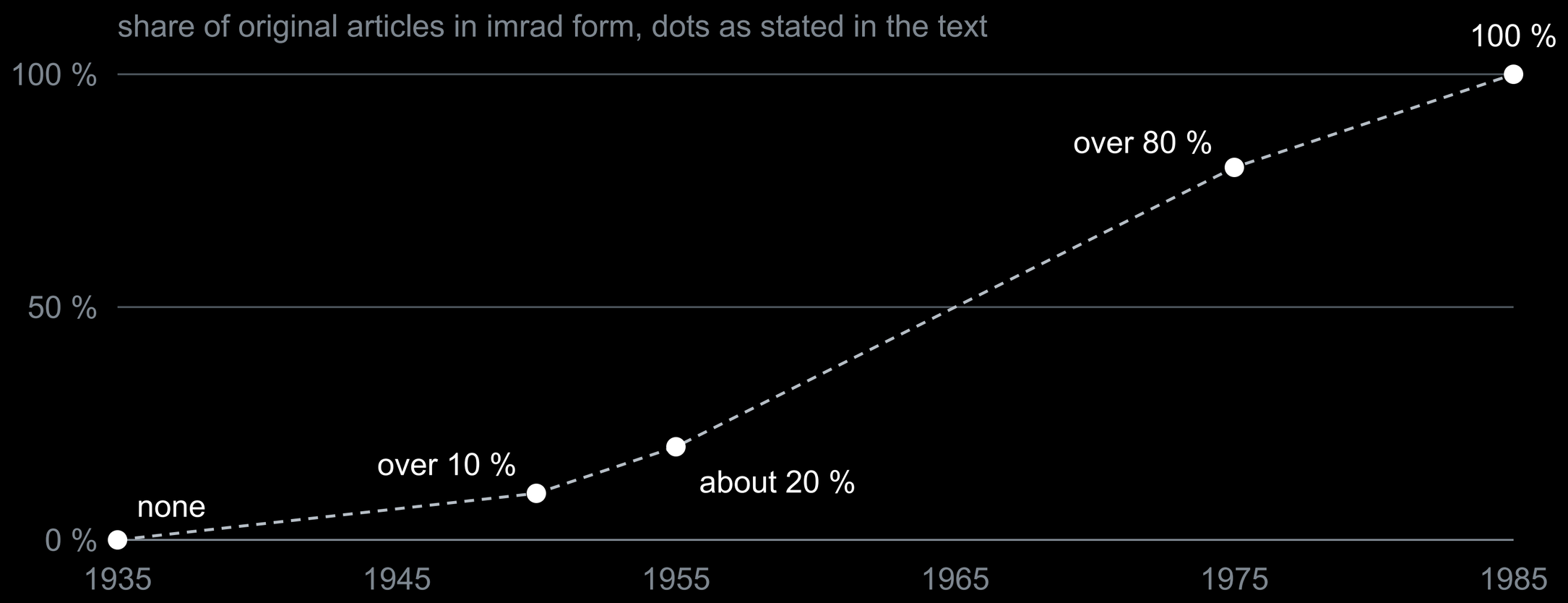
introduction	why, and what is the question?
methods	how did we find out?
results	what did we find?

references

title, abstract

introduction	why, and what is the question?
methods	how did we find out?
results	what did we find?
discussion	what does it mean?

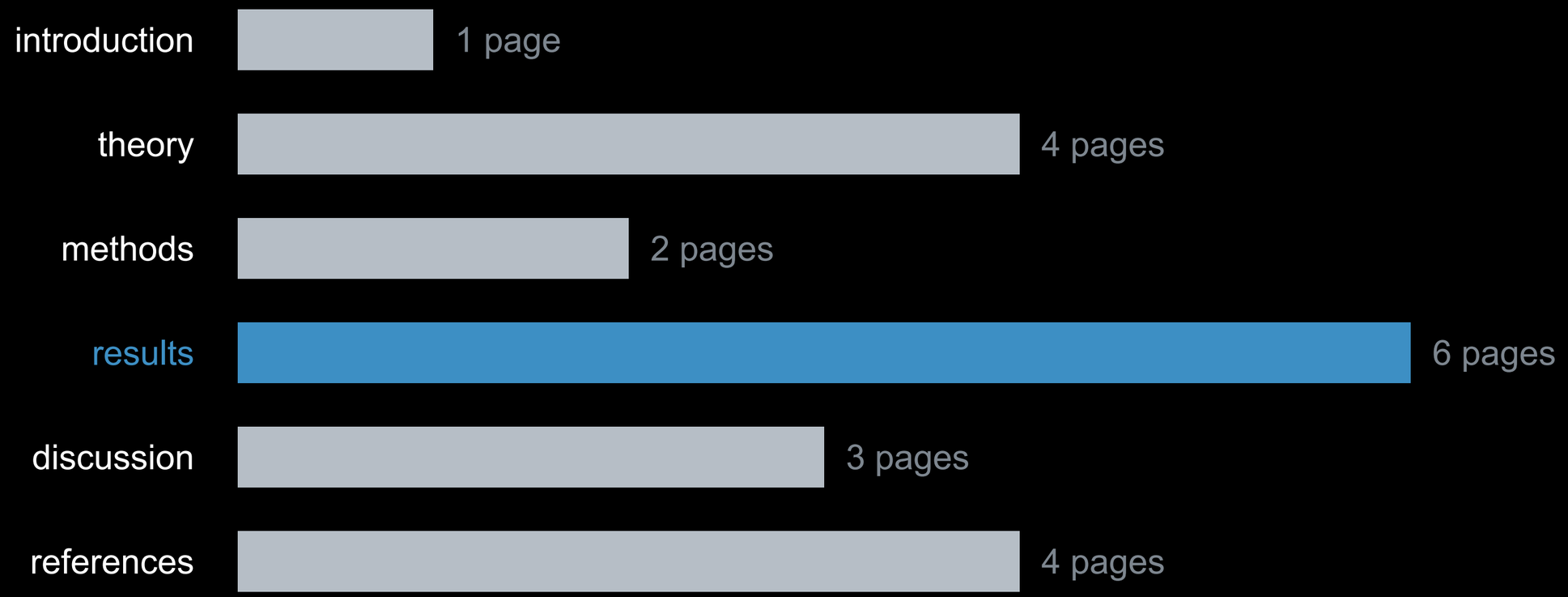
references



editors made imrad the rule, because readers can then jump straight to the section they need.

Sollaci & Pereira (2004), J Med Libr Assoc 92(3), 364–367: 1,297 original articles in BMJ, JAMA, The Lancet, NEJM

how long is each part?



a rule of thumb for an empirical paper of 20 pages

the results take the most room, and the theory must not crowd them out.

after Döring (2023), Forschungsmethoden und Evaluation, 6th ed., p. 771

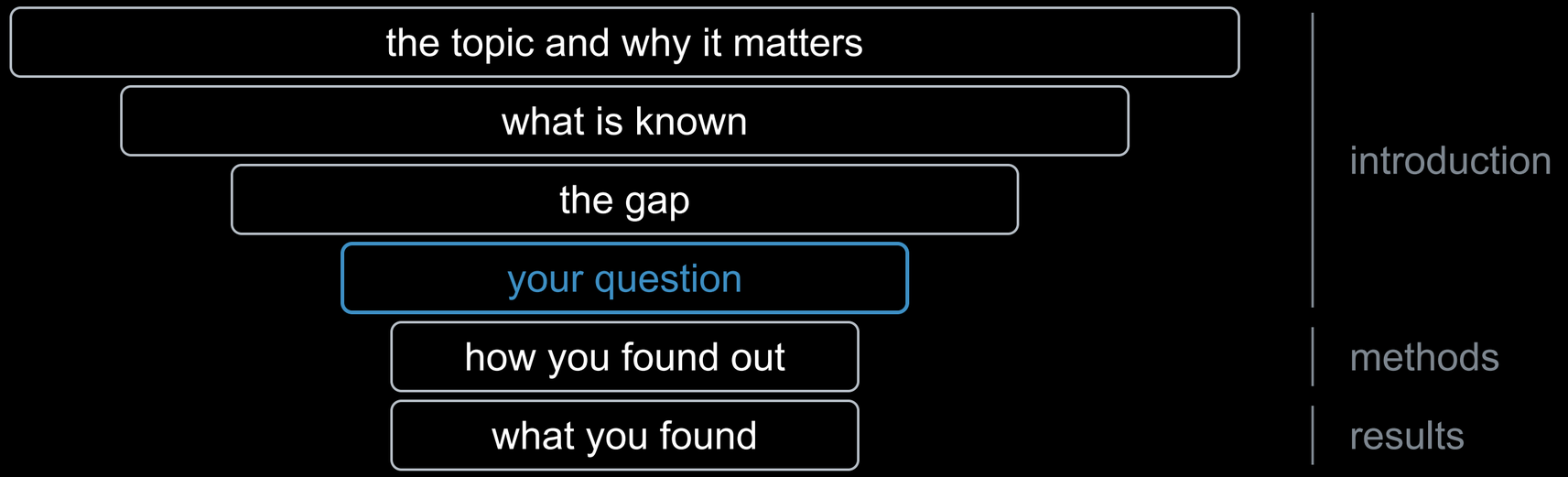
the hourglass



the introduction ends with your question, and the discussion begins with its answer.

after Cals & Kotz (2013), effective writing and publishing scientific papers, J Clin Epidemiol 66

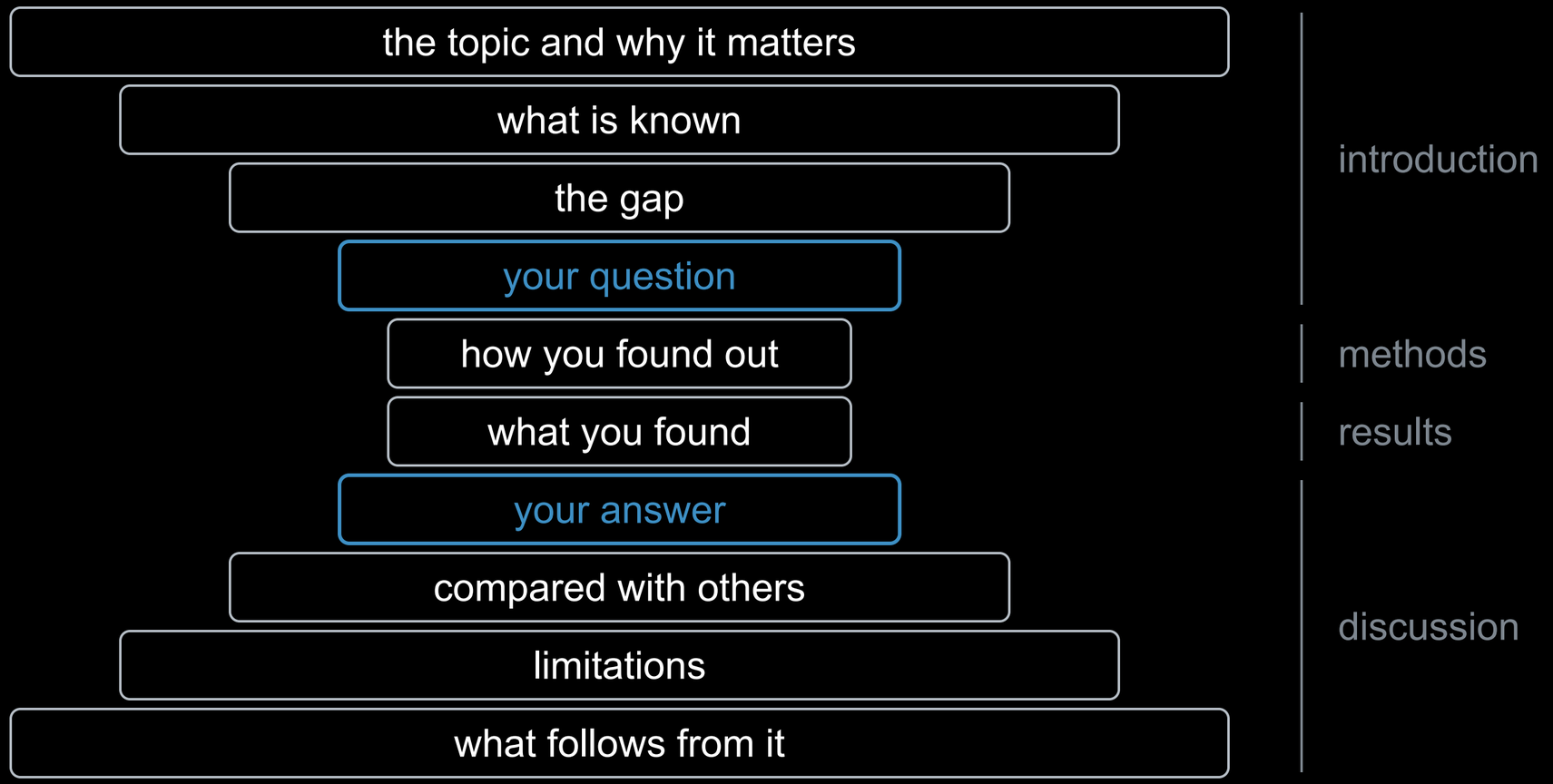
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part 2

what goes where

one job for each section

✓ goes in

why the topic matters

what is already known

the gap: what is not known yet

your question, in the last paragraph

stays out

a summary of the whole field

your results

a question that appears only later

the introduction narrows like a funnel from the topic down to your question.

after Cals & Kotz (2013), part III: introduction

✓ goes in

the data source, and that you reuse it

the sample and how it was recruited

the questions you use, word for word

how you cleaned the data

how you analysed them

stays out

what a mean or a median is

your results

the methods are a recipe that would let someone else repeat your analysis.

APA JARS-Quant (2020); Döring (2023), p. 771

✓ goes in

findings in the order of your questions

numbers, with the size of each group

tables and figures

each finding once: text, table or figure

stays out

why it is so

what it means

words like “remarkably”

the results report what the data say, and leave what it means to the discussion.

after Döring (2023), p. 772, and Cals & Kotz (2013), part V: results

✓ goes in

the answer to your question, first

how it compares with other studies

the limitations of data and method

what follows for practice and research

stays out

new results

every number once again

a vague “more research is needed”

the discussion opens with the answer and then widens like an upside-down funnel.

after Döring (2023), p. 770, and Cals & Kotz (2013), part VI: discussion

part 3

the most common mistake

interpreting in the results

where does this sentence belong? “vegetarians named a higher price limit, probably because they value animal welfare.”

introduction

methods

results

discussion



live.hsos.uber.space/q

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- introduction
- methods
- results
- discussion

live.hsos.uber.space/q

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- introduction
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- results
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live.hsos.uber.space/q

vegetarians named a higher price limit,
probably because they value animal welfare.

a result says what the data show, and the “because” waits for the discussion.

vegetarians named a higher price limit,
probably because they value animal welfare.

results

vegetarians named a higher price limit
than omnivores, with a median of
2.49 € against 1.99 €.

discussion

this may reflect that they value
animal welfare, as earlier studies
suggest.

a result says what the data show, and the “because” waits for the discussion.

health, environment or taste?

abstract
1. introduction
    1.1. theory of planned behaviour
    1.2. the current study
2. materials and methods
    2.1. procedure
    2.2. participants
    2.3. measures
    2.4. data analysis
3. results
    3.1. predicting intention
    3.2. predicting plant-based milk consumption
4. discussion
    strengths and limitations
5. conclusions
references

Dorina, Nikpour, Mullan & Uren (2025), Foods 14(11), 1970, CC BY 4.0: section headings of the article

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← the question, with hypotheses

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← the answer: first paragraph

← the limitations

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part 4

the abstract

the whole paper in one paragraph

the paper in miniature

problem milk competes on price, so it matters where people's price limits lie.

one sentence per part, about 150 to 250 words in all, written last

after Döring (2023), p. 776, and the APA publication manual (2020)

the paper in miniature

problem — milk competes on price, so it matters where people's price limits lie.

question — do people with different diets name different price limits for milk?

one sentence per part, about 150 to 250 words in all, written last

after Döring (2023), p. 776, and the APA publication manual (2020)

the paper in miniature

problem	milk competes on price, so it matters where people's price limits lie.
question	do people with different diets name different price limits for milk?
method	we reanalysed an existing survey; 611 of 2,809 people named a limit.

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question	do people with different diets name different price limits for milk?
method	we reanalysed an existing survey; 611 of 2,809 people named a limit.
findings	vegetarians' median limit was 2.49 €, that of omnivores 1.99 €.
conclusion	price limits may differ by diet, which a larger study should test.

one sentence per part, about 150 to 250 words in all, written last

after Döring (2023), p. 776, and the APA publication manual (2020)

write your question at the end of the introduction and your answer at the start of the discussion, and let everything else connect the two.



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